

Powder Particle

Powder particles produced by the calcium hydride process have a spongy morphology similar to sponge fines produced by the sodium process and consist of conglomerations of small particles fused together during the reaction process.

From: [Handbook of Non-Ferrous Metal Powders, 2009](#)

Related terms:

[Mechanical Alloying](#), [Sintering](#), [Microstructure](#), [Selective Laser Melting](#), [Binders](#), [Nozzle](#), [Melt Pool](#), [Porosity](#)

[View all Topics](#)

Material Characterization for Additive Manufacturing

Chee Kai Chua, ... Wai Yee Yeong, in [Standards, Quality Control, and Measurement Sciences in 3D Printing and Additive Manufacturing](#), 2017

4.1 Powder Size Measurements

Powder particle sizes have a direct influence on the layer thickness and minimum feature size of an AM part. Smaller powder particles permit a thinner layer thickness, finer minimum feature size and better surface finish. After a sample was collected the powder size distribution can be determined using one of the several methods mentioned below:

- microscopy,
- sieving,
- gravitational sedimentation, and
- light scattering.

The first method of measuring powder size is to use microscopy techniques, such as [optical microscopes](#), scanning [electron microscopes](#) and/or transmission electron

microscopes. These techniques allow one to directly see and measure the various dimensions of the powder particles.

The second method is to use a series of sieves with different mesh openings to separate the powder into the different sizes.

ASTM B214-07 Standard test method for sieve analysis of metal powders.

This test method covers the dry sieve analysis of metal powders or mixed powders, using sieves with size between 45 and 1000 μm .

The sieves are stacked from top to bottom in the order of decreasing mesh opening and a collection pan is placed below the entire series of sieves. The shaker, which the entire set up is attached to, runs for 15 minutes and induces the sieving action.

The third particle size determination method is based on gravitational sedimentation

ASTM B761-06 Standard test method for particle size distribution of metal powders and related compounds by X-ray monitoring of gravity sedimentation.

This test method is applicable to particles of uniform density and composition, having particle size distribution between 0.1 and 100 μm . The reported particle size measurement is a function of, both the actual particle dimension and shape factor, as well as the particular physical or chemical properties being measured.

In this method, the attenuation of a horizontally collimated beam of X-ray is measured after it passes through a liquid suspension containing powder particles of different sizes. Initially, the powder particles are homogeneously mixed via circulation and the attenuation of the X-ray is at a maximum. Once circulation ceases, all particles begin settling to the bottom with the larger particles sinking at a faster rate than the others, as shown in Fig. 5.16. Under the [laminar flow](#) regime, the [settling velocity](#) of the particles can be directly related to an equivalent [Stokes](#) diameter through the [Stokes equation](#). Therefore, particle sizes can be determined if the settling velocity of the particles can be obtained under a [low Reynolds number](#) flow. Since the vertical distance of the X-ray beam from the top of the cylinder is known, and the time taken for particles of different sizes to sink from the top of the cylinder to the X-ray beam can be obtained from the X-ray attenuation plot, the [particle velocities](#) can be computed. Consequently, the particle size distribution can also be obtained.

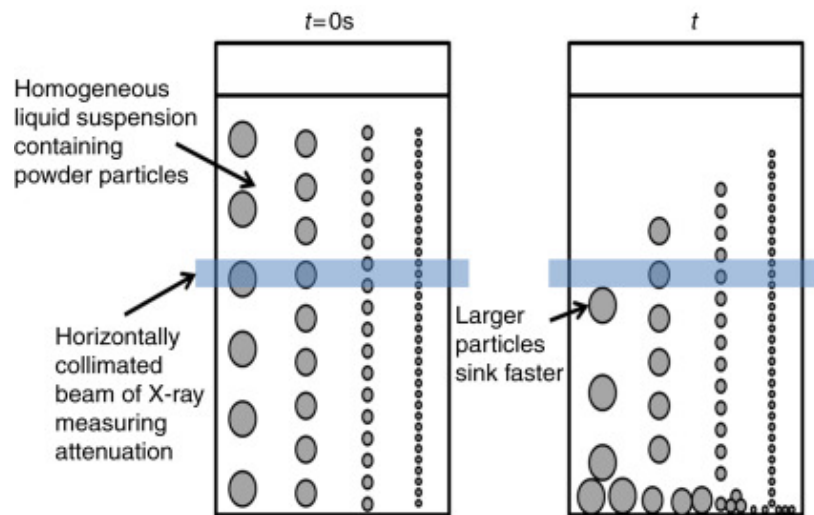


Figure 5.16. Illustration of Gravitational Sedimentation Method.

The last method measures size using the information contained in the light scattered from the powder particles.

ASTM B822-10 Standard test method for particle size distribution of metal powders and related compounds by light scattering.

This test method covers the determination of the particle size distribution by light scattering, reported as volume percent, of [particulate materials](#) including metals and compounds. It applies to analyse with both aqueous and non-aqueous dispersions. In addition, analysis can be performed with a gaseous dispersion for materials that are hygroscopic or react with a liquid carrier. The standard is applicable to the measurement of particulate materials in the range of 0.4–2000 μm , or a subset of that range, as applicable to the particle size distribution being measured.

Scattering can be achieved either by dispersing the particles in a liquid medium and circulating it through the light beam or by aspirating the dry particles in a carrier gas. The scattered light is captured by [photodetector](#) arrays, which then convert it to electrical signals for processing. By invoking the Mie Scattering or Fraunhofer Diffraction theory, or both, the collected signals can be converted into size distribution data.

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Debinding and sintering of metal injection molding (MIM) components

S. Banerjee, C.J. Joens, in [Handbook of Metal Injection Molding](#), 2012

7.4.8 Effect of powder size and surface area

The powder particle size has a direct relationship with the surface area of the powder. In order to understand the effect of surface area in MIM let us calculate the surface area of 1 g of powder, assumed to be made up of 100% of 10 particles, to simplify the calculations. Then,

where the density of carbonyl iron powder is 7.8 kg/m³.

Hence,

Thus, the surface area of 1 g of the 10 powder is the number of particles in the 1 g of powder times the surface area of one powder particle, which works out to be about 77.5 m².

To give a perspective, 1 g of 10 powder, which is a little over a pinch of powder, has a surface area of approximately 80 m² (850 ft²), the size of a small apartment. In spite of what the calculations show, this fact is difficult to either visualize or comprehend, compared to 1 g of iron as one spherical particle, which has a diameter of 6.25 mm and a surface area of 127.8 mm², or a square with sides measuring 11.3 mm only. In a typical *D*₉₀ 10 powder used in MIM the particles are finer and the surface area is even larger and this entire area is a lake of exposed metal atoms with this humongous surface energy. While this surface energy aids the [sintering process](#), this huge surface area also leaves the material prone to oxidation with exposure to the ambient atmosphere, so that a can of open powder would show increasing oxygen content with increasing time of exposure. For this reason, sintering atmospheres and hence the material of furnace construction play very important roles in the properties of the parts processed.

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Manufacturing of Parts From Tetrafluoroethylene Homopolymers

Sina Ebnesajjad, Pradip R. Khaladkar, in [Fluoropolymer Applications in the Chemical Processing Industries](#), 2018

5.1.2.2 Densification and Sintering Mechanism

Resin powder particles are separated by air; it must be removed during preforming and sintering. PTFE particles undergo changes from the compressed to sintered stages. The powder is charged to the mold, compressed, and held for a [dwell period](#). After the preforming cycle is completed the preform is removed from the mold and allowed to rest for stress relaxation and degassing.

The preform expands slightly during relaxation and recovery. The pressure applied to the molded resin causes various changes in the PTFE particles. [Resin particles](#) undergo [plastic deformation](#) and are intermeshed together leading to the development of cohesive or *green* strength. Particles also deform elastically and experience cold flow under pressure. The air trapped in the space between the resin particles is compressed. Removal of pressure allows the recovery of [elastic deformation](#), which creates a quick “snap back” of the preform. Over time, stress relaxation partly reverses the cold flow and the preform expands.

The trapped air in the preform, at the time of removal from the press, is under high pressure, which is roughly equivalent to the preforming pressure. The air requires time to leave the preform because it is mostly contained in the void areas surrounded by enmeshed particles. Immediate sintering would lead to a rise in the already high air pressure and catastrophic cracking of the part as the PTFE melts and the mechanical integrity decreases. The preform should be allowed to degas which equalizes the internal air pressure to atmospheric pressure.

Sintering of the preform takes place in an oven where massive volumes of heated air are circulated. Initial heating of the preform leads to thermal expansion (Fig. 5.4) of the part. After PTFE melts, relaxation of the residual stresses occurs (stored because of the application of pressure to the polymer) where additional recovery takes place and the part grows. The remaining air begins to diffuse out of the preform after heating starts. Adjacent [molten particles](#) begin to coalesce slowly; usually hours are required because of the massive size of PTFE molecules. Fusion of the particles is followed by elimination of the voids, where almost no air is left. Complete elimination of the voids in PTFE is quite difficult because of the limited mobility of the large [polymer molecules](#).

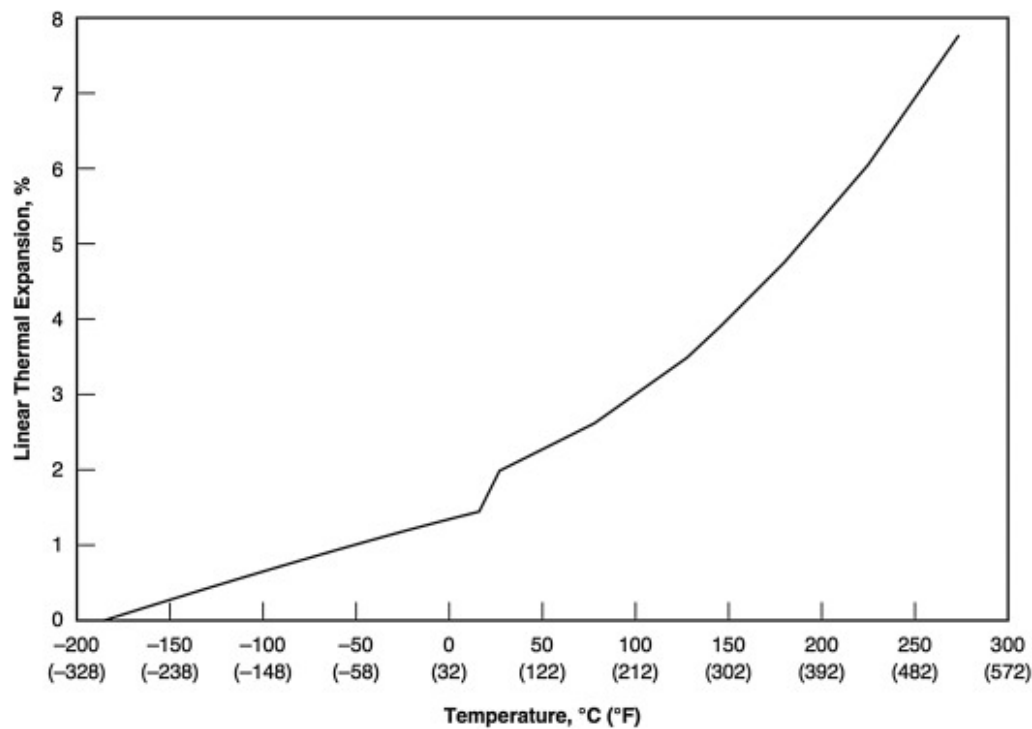


Figure 5.4. Transition point and linear thermal expansion of PTFE [4]. Courtesy: The Chemours Company.

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Fabrication and Processing of Granular Polytetrafluoroethylene

Sina Ebnesajjad, in [Fluoroplastics \(Second Edition\)](#), Volume 1, 2015

10.3.2 Densification and Sintering Mechanism

Resin powder particles are separated by air which is removed during preforming and sintering. Figure 10.8 depicts the changes that PTFE undergoes from the powder stage to a [sintered part](#). The powder is charged to the mold and compressed and held for a [dwell period](#). After the preform is made, it is removed from the mold and allowed to rest for stress relaxation and degassing.

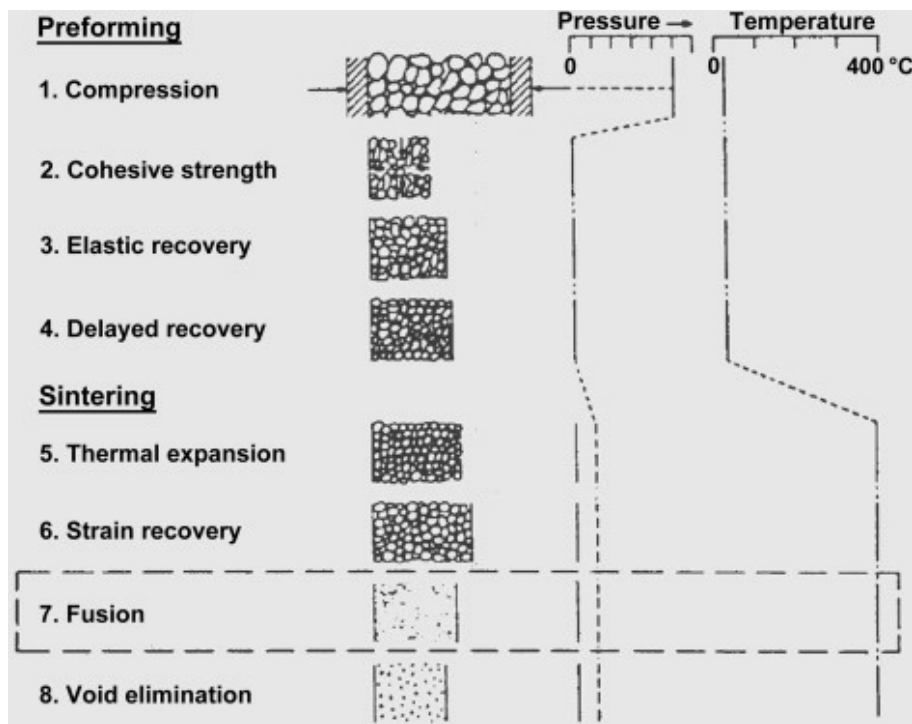


Figure 10.8. Schematic diagram of preforming and sintering sequence with polytetrafluoroethylene.²

The preform expands due to relaxation and recovery in steps 3 and 4 in Fig. 10.8. The pressure placed on the molded resin exerts three types of changes in the particles of resin. Resin particles undergo plastic deformation and are intermeshed together leading to the development of cohesive or *green* strength. Particles also deform elastically and experience cold flow under pressure. The air trapped in the space between the resin particles is compressed. Removal of pressure allows the recovery of elastic deformation, which creates a quick “snap back” of the preform. Over time, stress relaxation partly reverses the cold flow, and the preform expands.

The trapped air in the preform is under high pressure, theoretically equal to the preforming pressure. The air requires time to leave the preform because it is mostly contained in the void areas surrounded by enmeshed particles. Immediate sintering would lead to a rise in the already high pressure of the air and catastrophic cracking of the part as PTFE melts and the mechanical strength declines. The preform should be allowed to degas to equalize the internal air pressure to the atmospheric pressure.

Sintering of the preform takes place in an oven where massive volumes of heated air are circulated. Initial heating of the preform leads to thermal expansion (Fig. 10.8) of the part. After PTFE melts, relaxation of the residual stresses occurs (stored because of the preforming pressure) where additional recovery takes place and the part grows. The remaining air begins to diffuse out of the preform after heating starts. At the sintering temperature, adjacent melted PTFE particles fuse together and coalesce. The adjacent molten particles begin to fuse together by forming a neck followed by coalescence (Fig. 10.9). After two particles have completely coalesced,

they are indistinguishable from a larger particle. Voids are eliminated under the driving force of surface tension and elastic deformation.^{6,7} Smaller particle resins and higher preform pressures improve coalescence.

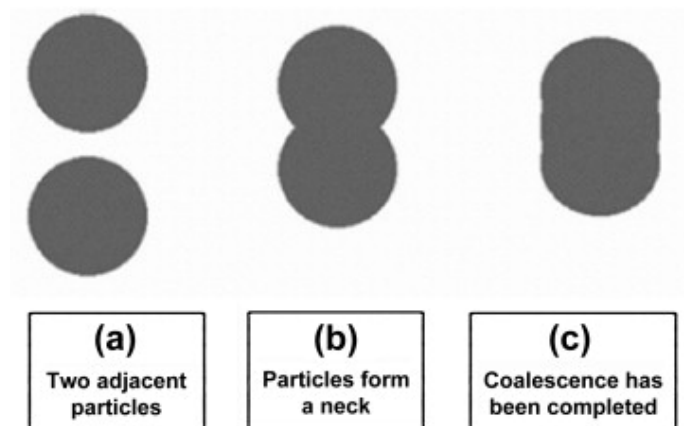


Figure 10.9. Mechanism of the sintering of PTFE.

Figures 10.10 and 10.11 shows coalescence of PTFE particles in a laboratory simulation of sintering after 100, 1000, and 10,000 s hold time above its melting point. Usually hours are required because of the massive size of PTFE molecules (molecular weight is 10^6 – 10^7). Fusion of the particles is followed by elimination of the voids, where almost no air is left. It is noteworthy to remember that the elimination of all the voids in PTFE is quite difficult because of limited mobility of the large [polymer molecules](#).

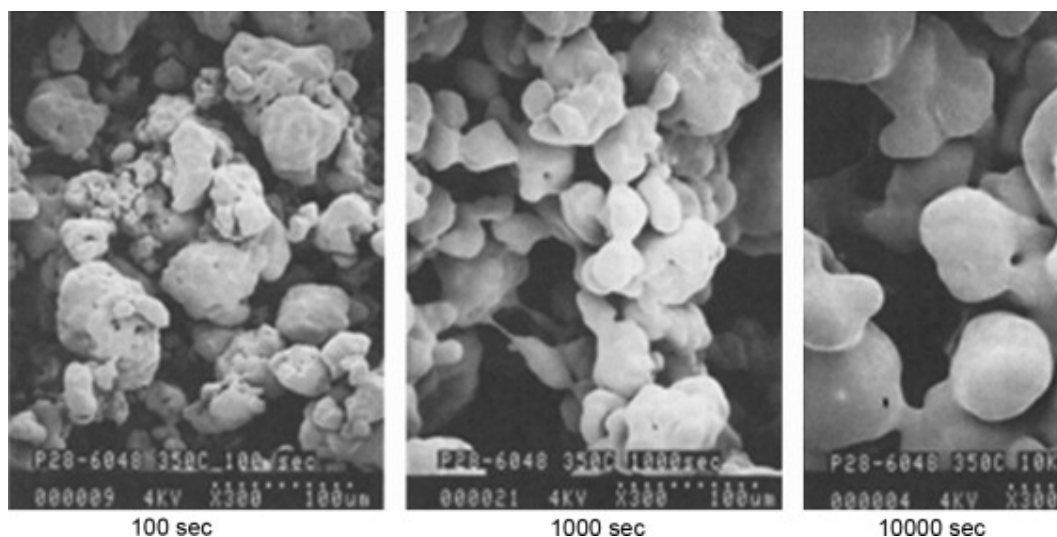


Figure 10.10. Micrographs (300x) showing evolution of the coalescence of PTFE particles over time.⁷

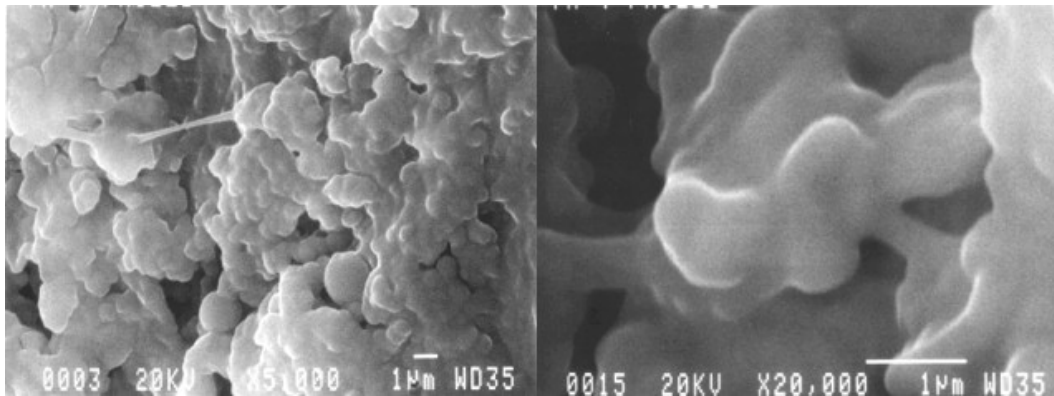


Figure 10.11. Micrographs (5000x and 20,000x) showing the coalescence of PTFE particles over time.⁷

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Nickel-based Superalloys: Alloying Methods and Thermomechanical Processing

J.H. Weber, in [Encyclopedia of Materials: Science and Technology](#), 2001

2.2 Consolidation

Compacting the [powder particles](#) into solid forms involves three steps: [encapsulation](#) of the powder in a container; evacuation of the sealed container to remove contained gas; and the [compaction process](#) itself. The container for the powders is normally made of mild steel, but [glass and ceramic](#) containers are also used. After sealing, the containers are evacuated to remove interparticle gases and reduce particle [surface contamination](#). Evacuation may be done at either ambient or elevated temperatures.

The most common consolidation practice for [superalloy](#) powders is [hot isostatic pressing](#) (HIP). This is performed in an autoclave, a large [pressure vessel](#), using isostatic gas pressure to densify the powder into a solid. The control parameters during [HIP](#) are the temperature, gas pressure, and process time. The consolidated form is dependent upon the shape and design of the [powder container](#). Shapes range from simple round cylinders to relatively complex three-dimensional disk configurations. Canned powders are also consolidated by [hot extrusion](#) and forging. The advantage of these mechanical working–consolidation practices is better interparticle bonding due to particle [deformation and surface shearing](#). Extrusion and forge consolidation is typically limited to relatively simple shapes.

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Ball and Rod Mills

Jean-Paul Duroudier, in [Size Reduction of Divided Solids](#), 2016

3.5.5 Cushion effect in dry milling

Look at the ultrafine particles, that is those particles with a diameter less than 10 μm . Here, we call these particles microscopic particles.

With such particles, powders obtained have a very high porosity. In fact, particles line up in more or less ramified chains and are more or less interconnected. The resultant effect resembles a steel frame where all the iron has been arranged haphazardly. We understand that the void fraction in volume for such a structure is high (0.7 or 0.8), while the void fraction for loose spherical balls is 0.4. The structure of powders is stable and does not collapse because the bond between the two particles is quite strong to mainly neutralize the weight of these particles. The force of the interparticle bond is due to van der Waals forces that we express quantitatively using Hamaker's law (see section 6.3.1 in [DUR 16]).

When a grinding body (ball) falls on a powder bed, the latter compresses by absorbing a significant amount of the ball's kinetic energy. This lost energy was dispersed in a volume that is much greater than the tiny hemisphere that included the stress in a plane in contact with a sphere. A hit particle could move back and disperse its kinetic energy in a powder bed.

When particles to be milled are made up of a mass greater than 15% of microscopic powder, the probability of fragmentation by collision or crushing could decrease dramatically.

Then, when a grinding body arrives on a powder bed, it destroys the powder's structure by compressing it like a sponge and, as a result, the interstitial air is expelled. But the permeability of a set of microscopic particles is very weak and air trying to leak into this environment brings particles from the structure with it, whence it is likely to release a cloud of dust.

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Electrostatic Powder Coating

III Back-Ionization

Even if [powder particles](#) are successfully deposited on a grounded workpiece, the overall performance of the coating system and the quality of the coating are still not guaranteed. After deposition, mechanical disruption of the uncured layer can seriously affect the evenness of the coating. This is a direct result of [electrical breakdowns](#) occurring through the layer thickness. The energy associated with such breakdowns not only disrupts the powder layer, but also seriously affects the charging efficiency of the powder applicator.

This important phenomenon is known as back-ionization, or back-discharge. An appreciation of the basic mechanisms and their implications on coating behavior is important. Figure 5 illustrates this rather complex phenomenon schematically.

FIGURE 5. Schematic of back-ionization in a deposited powder layer.

Three species of particles will alight on a workpiece when a standard corona-charged powder gun is used. These are charged particles, uncharged particles, and [negative ions](#). Since the powder particles generally are highly insulating, charge accumulation occurs within the bulk of the deposited powder layer. This charge includes that carried by the particles themselves, but most is the contribution presented by the arrival of [free ions](#) on the substrate. With this rapid growth in charge density, the voltage across the layer thickness increases correspondingly. Since [particle packing](#) within the layer is relatively loose, the [dielectric strength](#) of the layer is dictated primarily by the breakdown strength of the air channels between the particles. The layer breakdown illustrated in Fig. 5 is, in fact, electrical breakdown of air, being $3 \text{ MV} \cdot \text{m}^{-1}$ at NTP (normal temperature and pressure). This condition is reached very rapidly, normally within about 1 sec of coating time in a [high-voltage](#)

corona-charging system. In many instances, this may correspond to the deposition of the first monolayer of powder particles.

The energy associated with such a discharge can have serious disruptive effects on the deposited layer. The most common effect is the creation of small [craters](#) on the surface and, in some cases, complete puncturing of the layer, which results in discrete [pinholes](#) that penetrate through to the substrate. Heat curing of the powder can alleviate the effects of such damage. Appropriate [flow characteristics](#) in the paint when it is molten eliminates most pinholes so that the unevenness of the cratered surface may not be obvious. Often, however, some pinholes remain, and curing a cratered surface results in a finish resembling the texture on the skin of an orange. Such an orange-peel effect is highly undesirable when high-quality finishes are sought. Furthermore, the persistence of pinholes contributes to the early onset of substrate corrosion.

In addition to these two serious disruptive elements on the coating itself, perhaps an even more serious by-product of back-ionization is the fact that ions of opposite sign to the charging polarity are created. For example, in Fig. 5, the gun-charging polarity is shown as negative. Back-ionization generates both negative and [positive ions](#). The newly created negative ions are collected by the substrate, but the positive ions move in the opposite direction to the oncoming charged particles. In practice, the onset of back-ionization has been shown to annihilate particle charging, which ultimately leads to a collapse of the [electrostatic deposition](#) process.

The photograph in Fig. 6 illustrates the degree of back-ionization occurring on a small test plate when it is coated with a negatively charged powder by using a high-voltage [corona](#) gun. Each point of light emission is a layer discharge similar to that depicted in Fig. 5. This photograph was taken using an [image intensifier](#) so as to enhance the brightness of the discharges. However, it is likely that such a degree of back-ionization would be visible to the naked eye if it were viewed in total darkness.

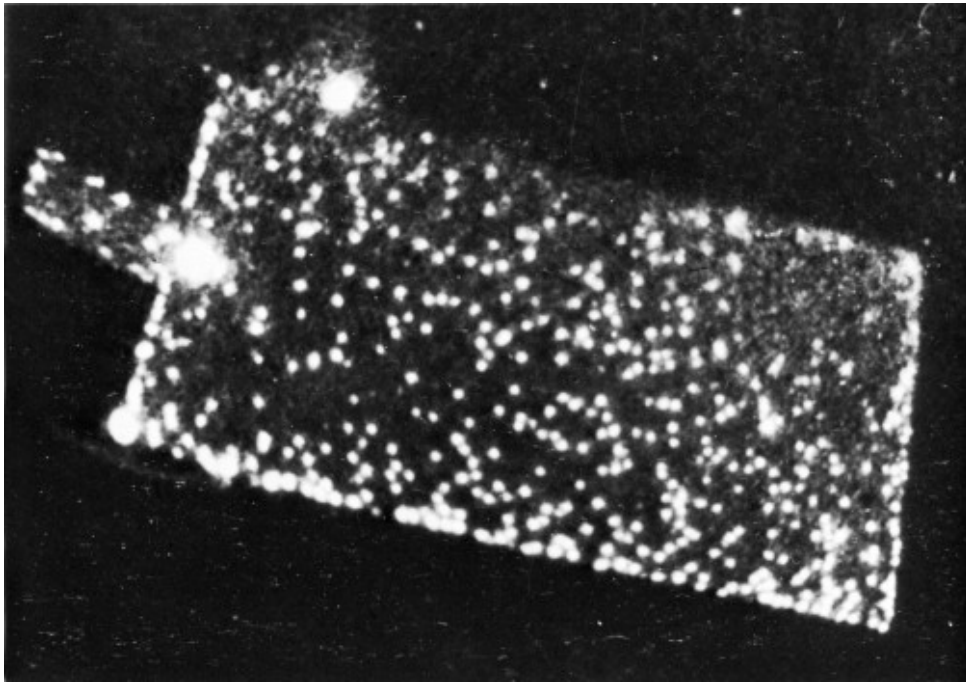


FIGURE 6. Image intensifier photograph of back-ionization on a powder-coated plate.

If the back-ionization process were viewed in real time, the corners and edges would be the first areas to exhibit back-ionization. An intensification of the electric field occurs in these areas, which results in a more rapid accumulation of powder there than on the flat surfaces and hence an earlier onset of back-ionization.

Another interesting characteristic of back-ionization is the apparent difference in layer disruption between positive and negative sign of powder charging. The reasons for this are complicated and will not be dealt with here, but an appreciation of the practical implications is important. For a more detailed explanation, refer to the sources listed in the Bibliography at the end of this article.

If the powder is charged negatively, the layer discharge is similar to that depicted in Fig. 5. That is, the discharge penetrates through the layer to the substrate. The discharge channel creates a pinhole in the powder layer, together with considerable disruption of the particle packing in the outermost layers. In addition to a deterioration in the quality of the finish, therefore, there will also be discrete [discontinuities](#) in the layer where moisture can easily penetrate, which will eventually lead to substrate corrosion. The protective qualities of the powder layer are therefore seriously degraded once negatively charged back-ionization is initiated.

Reversing the sign of the powder charging to positive has an interesting effect. Back-ionization still occurs, but the discharge is now a surface phenomenon. There will be no [complete penetration](#) of the powder layer, but surface-layer disruption will be more pronounced.

An important practical implication arises from this behavioral difference between negative and positive charging. If substrate protection is of prime importance, and quality of finish is not paramount, positive powder charging should be the operating polarity. Such requirements are often sought for the underside of vehicles, the insides of pipes and storage tanks, and so forth. However, if the quality of the finish is important, and protection is secondary, then negative powder charging should be used. This is often the requirement sought for domestic appliances, furniture, and decorative accessories.

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Design and Characterization of Nanoparticulate Drug Delivery

Basavaraj K. Nanjwade, ... Teerapol Srichana, in [Characterization and Biology of Nanomaterials for Drug Delivery](#), 2019

6.1.3 Bulk Density

The mass of the particle powder divided by the volume of the powder in a measuring cylinder is known as bulk density. In measurement of bulk density, voids between the particles and voids present within particles are considered. There are two types of density: initial bulk density and tapped density. Initial bulk density is one when the powder is transferred to a measuring cylinder, and tapped density is after continuous tapping of the measuring cylinder. The ratio of tapped density to the initial bulk density is known as the Hausner's ratio, which is used in measurement of powder compressibility and flowability.

[> Read full chapter](#)

Powder Characterization

K.S. Narasimhan, in [Encyclopedia of Materials: Science and Technology](#), 2001

3 Surface Area

The surface area of [powder particles](#) can be measured by the [permeation](#) technique or by the BET (Braunauer–Emmett–Taylor) method, which measures gas adsorption on powder surfaces. Specific surface area is calculated from the required amount of [inert gas](#) to cover the surface and the area of the molecule of adsorbate. The specific

surface areas of water atomized and sponge powders are roughly $0.1\text{--}0.12\text{ m}^2\text{g}^{-1}$ and $0.1\text{--}0.2\text{ m}^2\text{g}^{-1}$, respectively. Although surface area is a useful tool, it is rarely used in the characterization of metal powders.

Particle shape, size and distribution are important characteristics that define powder. These properties affect various other parameters used in the [compaction process](#) for the production of powder metal parts. Discussed below are powder characteristics that are most utilized in the P/M industry.

[> Read full chapter](#)

Carbonyl Method of Metal Powder Production

Irina V. Frishberg, in [Handbook of Non-Ferrous Metal Powders](#), 2009

Properties and Applications of the Metal Carbonyl Powders

The size of carbonyl powder particles varies in a broad range from 0.5 to $20\text{ }\mu\text{m}$. The features of these powders are high specific surface and considerable defects of crystalline structure; they are more active as compared to the powders manufactured by other techniques. However, these powders are not inflammable due to absorbed carbon monoxide that passivates the particle surface. The particle shape is close to spherical. Sometimes particles have a laminated 'onion' structure, imparting specific electromagnetic properties to the material.

Carbonyl powders can contain $1\text{--}3\%$ of carbon and oxygen (that is equivalent to about 30% of metallic compounds) while the amount of other impurities does not exceed $1 \times 10^{-3}\%$.

The powders can be subjected to additional treatments with the purpose of bringing them into accordance with the requirements relating to the purity and the particle size. Mechanical methods (crushing, mixing, sieving, separating and trapping) and thermal treatment are in use. The latter may cause changes in the crystalline structure of particles, thus enabling production of powders with targeted properties, e.g. light powders with coarse and porous particles, as well as to increase the metal content of the powder up to 99.96% as a result of the elimination of carbon and oxygen.

Carbonyl powders of metals, alloys and composites are used for manufacturing pure malleable metals, precision alloys, various solid and porous items (parts, strips, rods, filters, etc.), in electric and radio engineering, mechanical engineering, automation and other areas.

Detailed characteristics of powders are given for each metal in the corresponding chapters of Section 4.

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